



Lesson 5.1 — Introduction to Root Functions

Big idea: A root function takes the n th root of its input. Square roots restrict the domain (no negatives); cube roots accept everything. The shape and domain depend entirely on whether n is even or odd.

Quick review

Notation: n th root of $x = x^{1/n}$.

Even root ($n = 2, 4, \dots$): domain $x \geq 0$, range ≥ 0 . Graph is a half-curve from the origin.

Odd root ($n = 3, 5, \dots$): domain all reals, range all reals. Graph passes through origin and extends in both directions.

Inverses: $x^{1/n}$ is the inverse of x^n on the appropriate domain.

Worked example

Worked example. Compare $f(x) = \sqrt{x}$ and $g(x) = \sqrt[3]{x}$.

f : domain $[0, \infty)$, range $[0, \infty)$. g : domain $\mathbb{R}$, range $\mathbb{R}$. Both pass through $(0, 0)$ and $(1, 1)$. g also passes through $(-1, -1)$.

Warm-ups

1. Compute $\sqrt{64}$.
2. Compute $\sqrt[3]{-27}$.
3. Compute $\sqrt[4]{-16}$. (Real number?)

Practice

1. State the domain of $f(x) = \sqrt{x - 5}$.
2. State the domain of $g(x) = \sqrt[3]{x + 2}$.
3. Find $f(0)$, $f(1)$, $f(4)$ for $f(x) = \sqrt{x}$.

Challenge

1. For what real values is $\sqrt{x - 1} + \sqrt[3]{x - 1}$ defined?
2. Sketch (describe) $f(x) = \sqrt{x}$ and $g(x) = \sqrt[3]{x}$ on the same axes. State three differences in shape or domain.



Answer key — Lesson 5.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 8.
2. -3 .
3. Not real (no real number squares to a negative).

Practice

1. $x - 5 \geq 0 \Rightarrow x \geq 5$. Domain: $[5, \infty)$.
2. All reals (cube root accepts negatives).
3. 0, 1, 2.

Challenge

1. Need $x - 1 \geq 0$ (for $\sqrt{}$) and (always for $\sqrt[3]{}$). Combined: $x \geq 1$.
2. Differences: (1) $\sqrt{x}$ is only defined for $x \geq 0$; $\sqrt[3]{x}$ is defined for all x . (2) $\sqrt[3]{x}$ has odd symmetry about the origin; $\sqrt{x}$ has no symmetry. (3) $\sqrt{x}$ rises and is concave down; $\sqrt[3]{x}$ is increasing on all reals with an inflection at the origin.